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(54) **VENTILATED HEADWEAR WITH
LOUVERED CONSTRUCTION**

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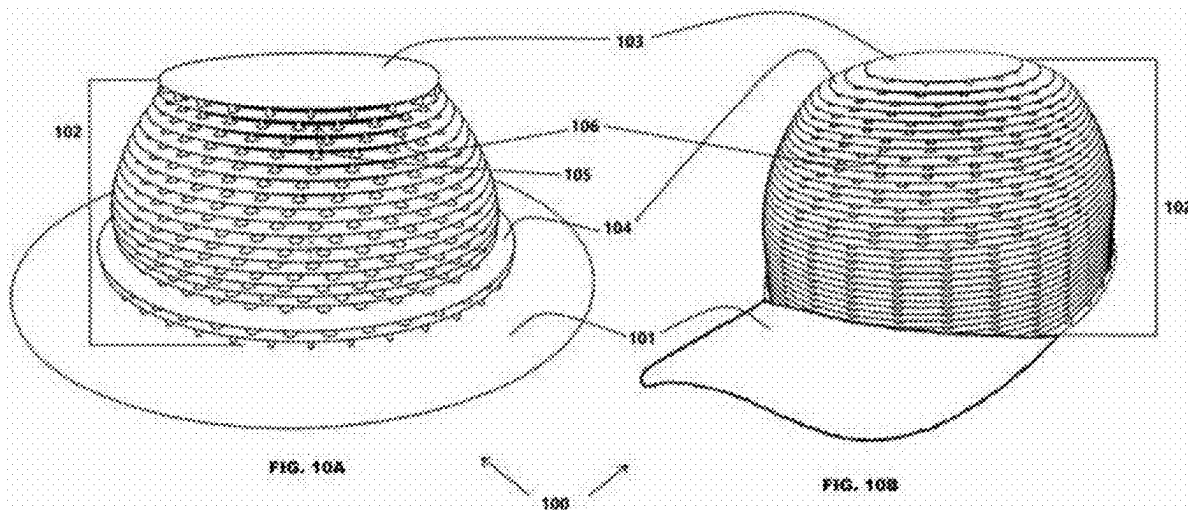
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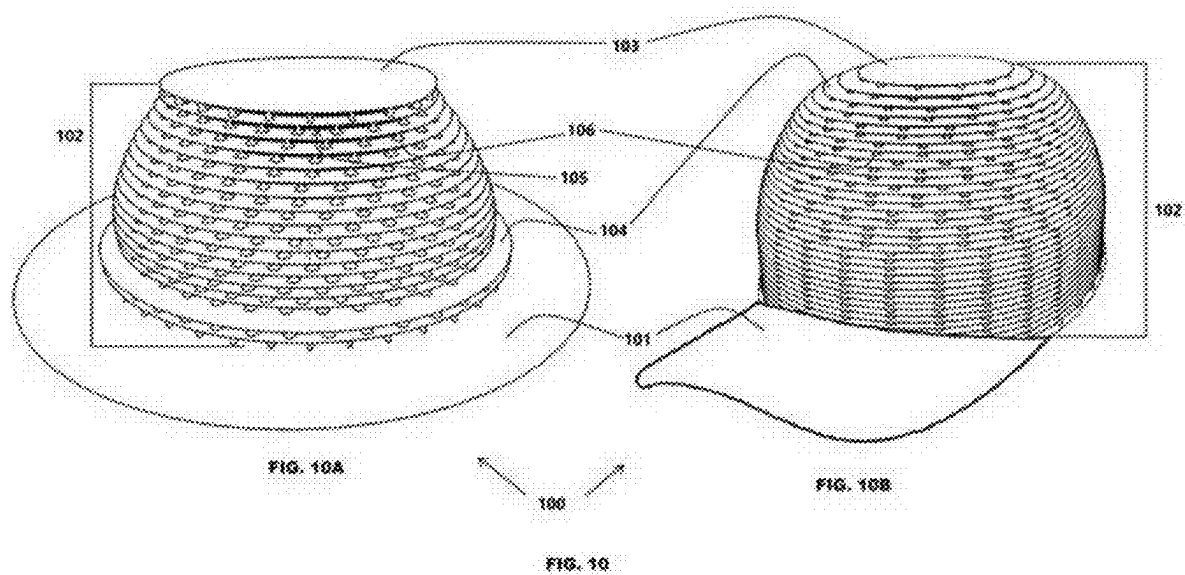
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(57) **ABSTRACT**

Headwear is disclosed that provides both shade and enhanced ventilation. The headwear comprises a plurality of horizontally-arranged, vertically-spaced louvers forming a hat body that encircles the wearer's head. The louvers are configured with a defined shade angle to block direct sunlight while allowing unimpeded airflow through the spaces between the louvers. Spacers maintain the vertical separation between adjacent louvers. This louvered construction provides significantly improved cooling compared to conventional hats, such as baseball caps or hats with mesh panels, which either trap heat or fail to provide adequate shade. The headwear can be manufactured using additive (3D printing) or subtractive (cutting and assembly) techniques, utilizing materials such as plastics or composites. A top member encloses the uppermost louver. A brim may optionally be included.





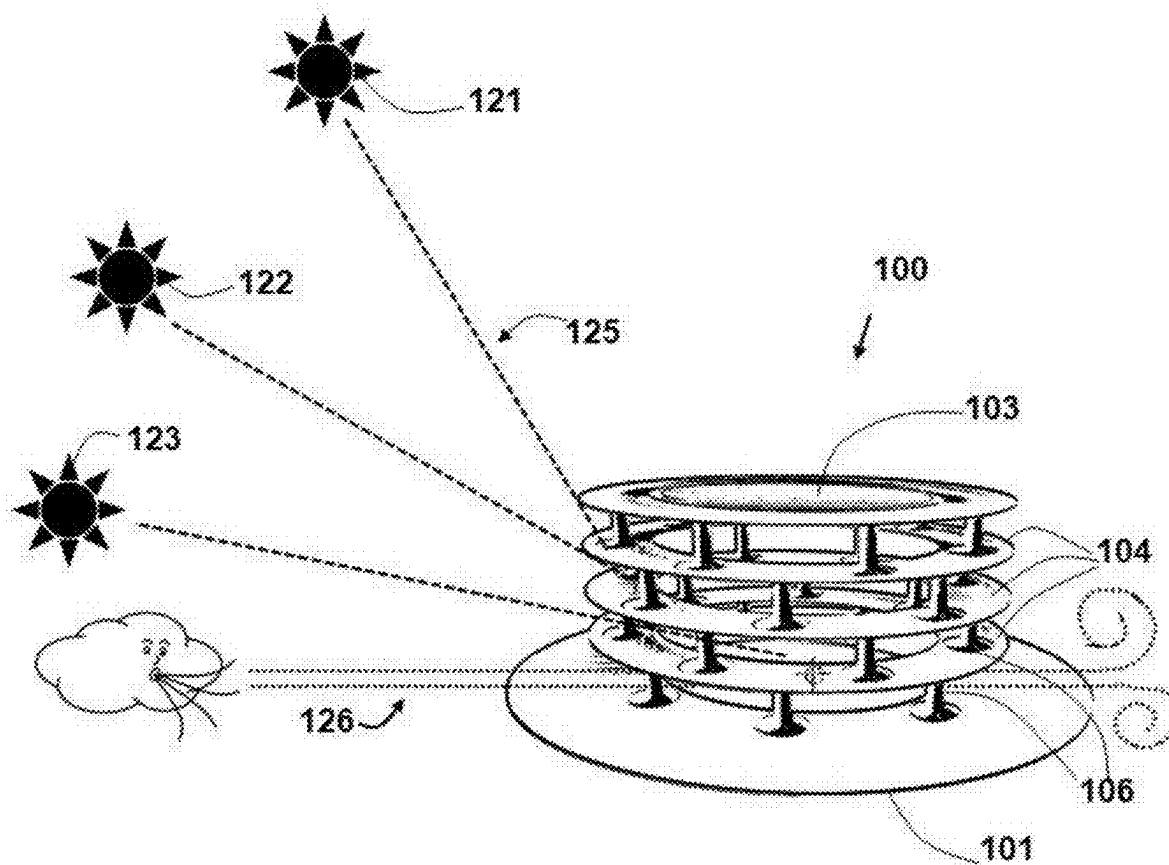
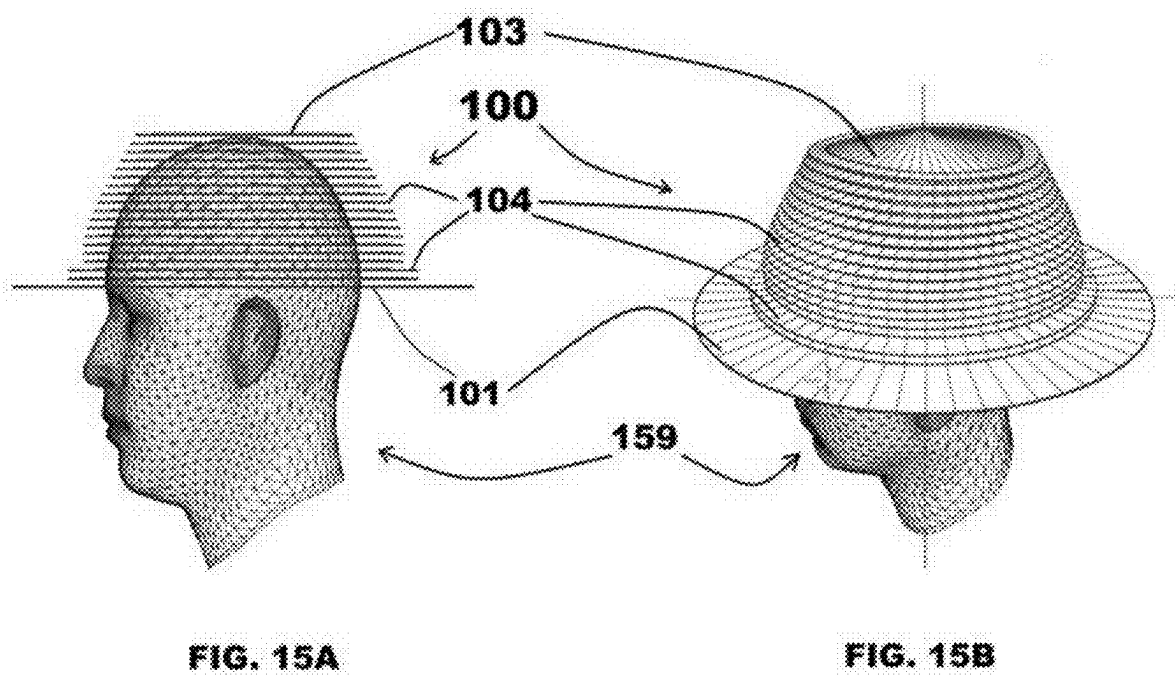


FIG. 12



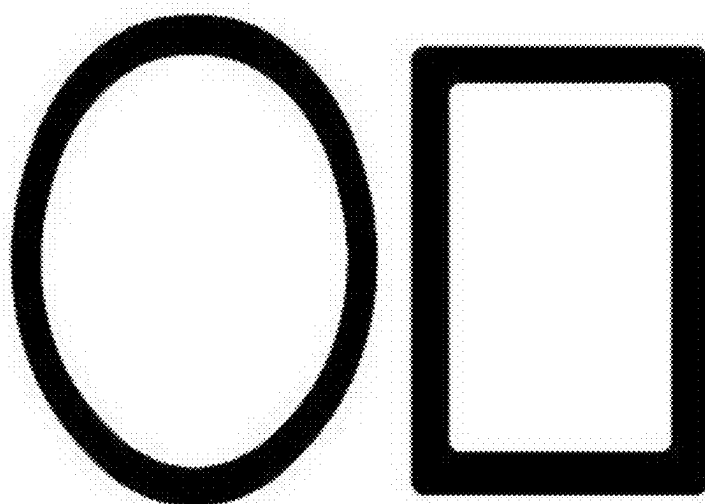


FIG. 20A

FIG.
20B

FIG. 20

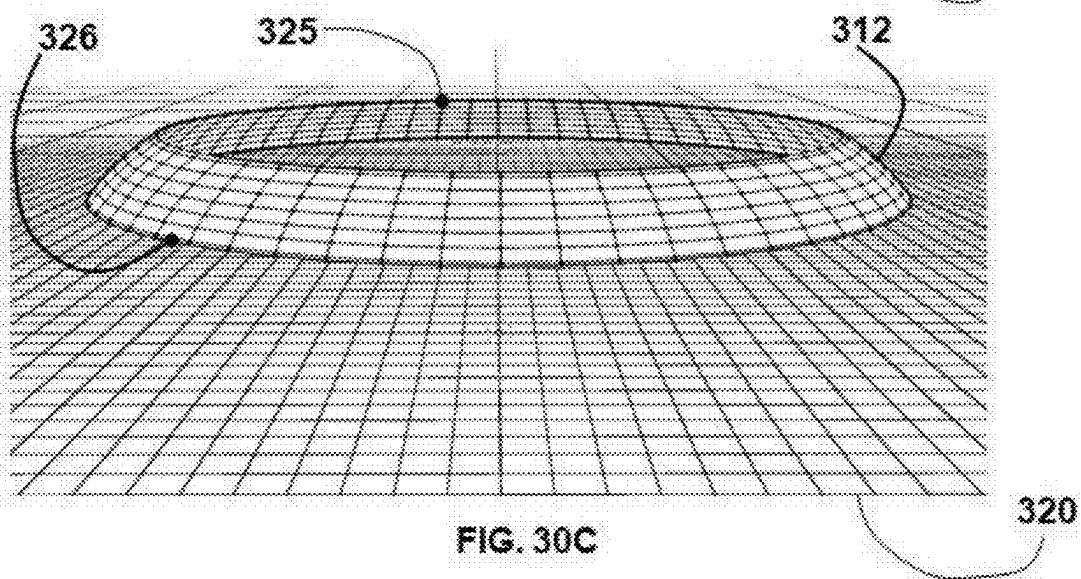
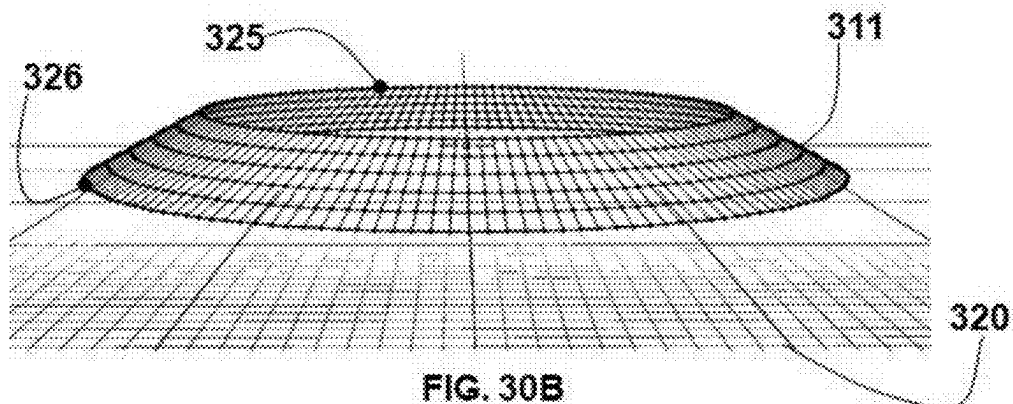
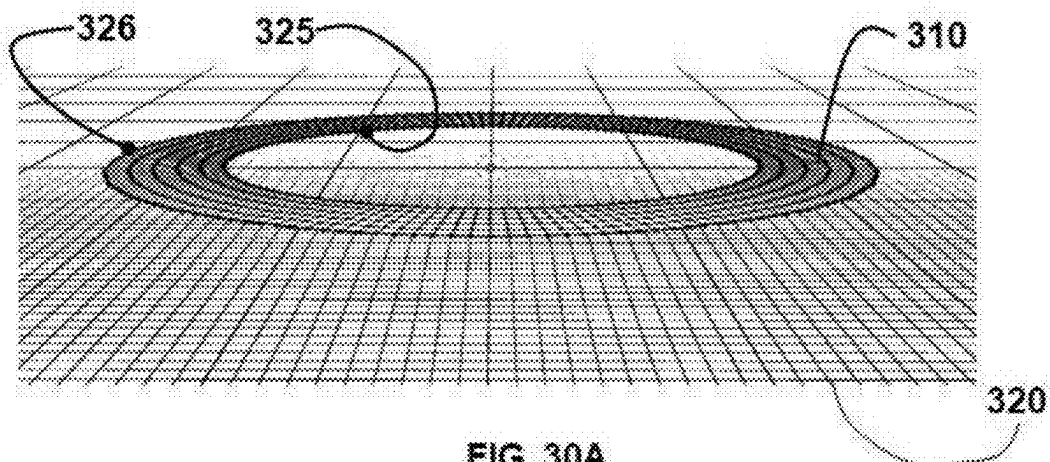


FIG. 30

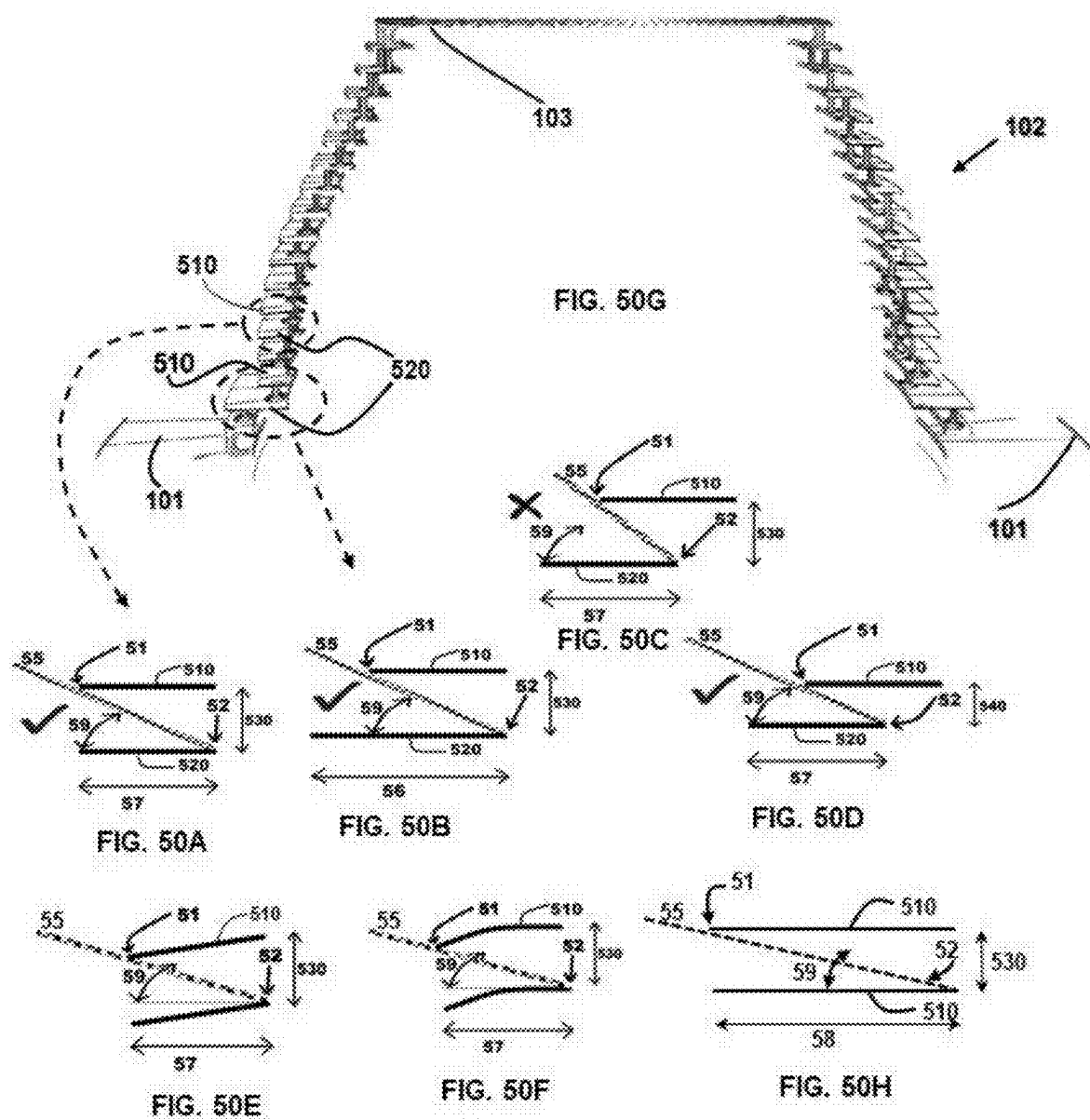


FIG. 50

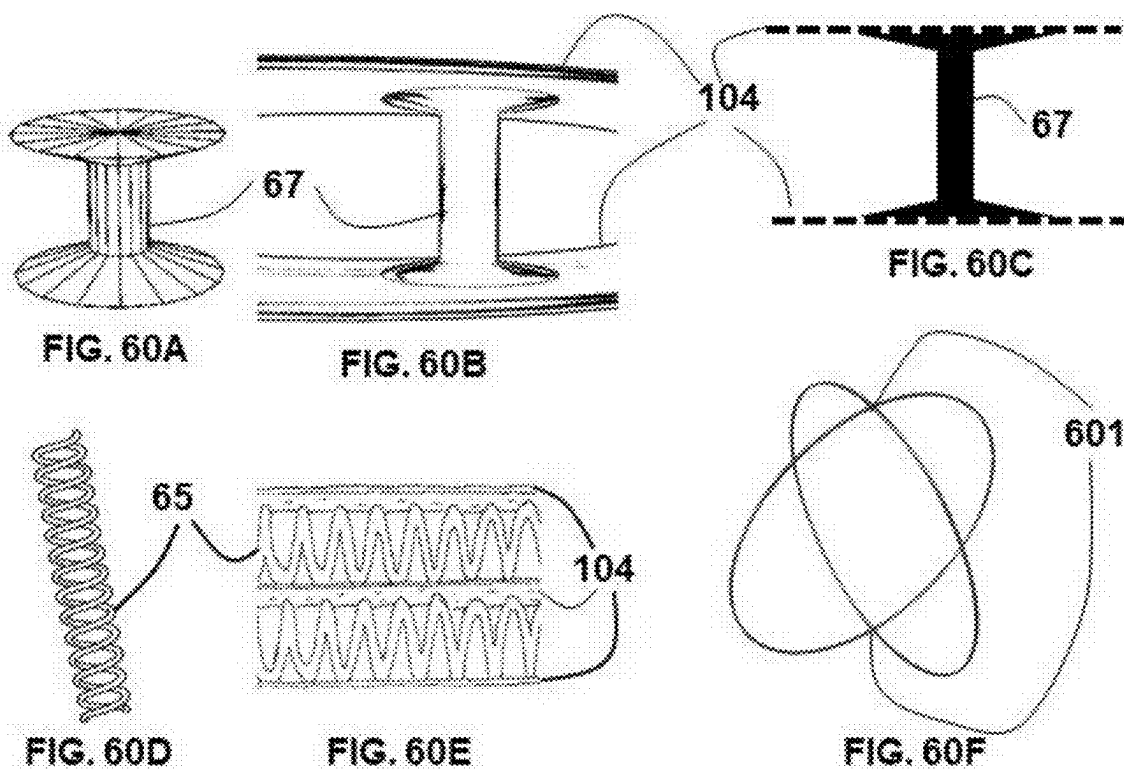
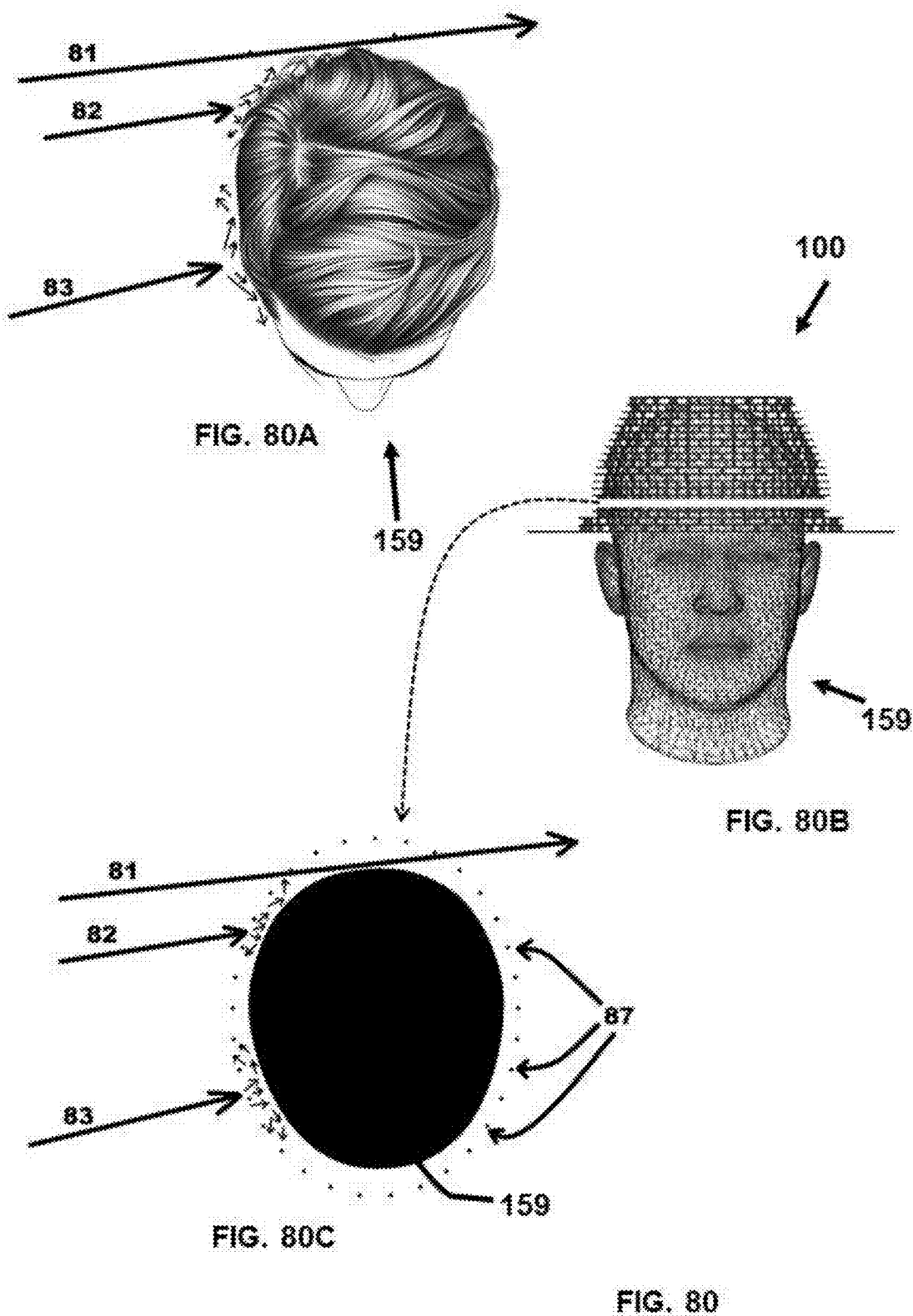


FIG. 60



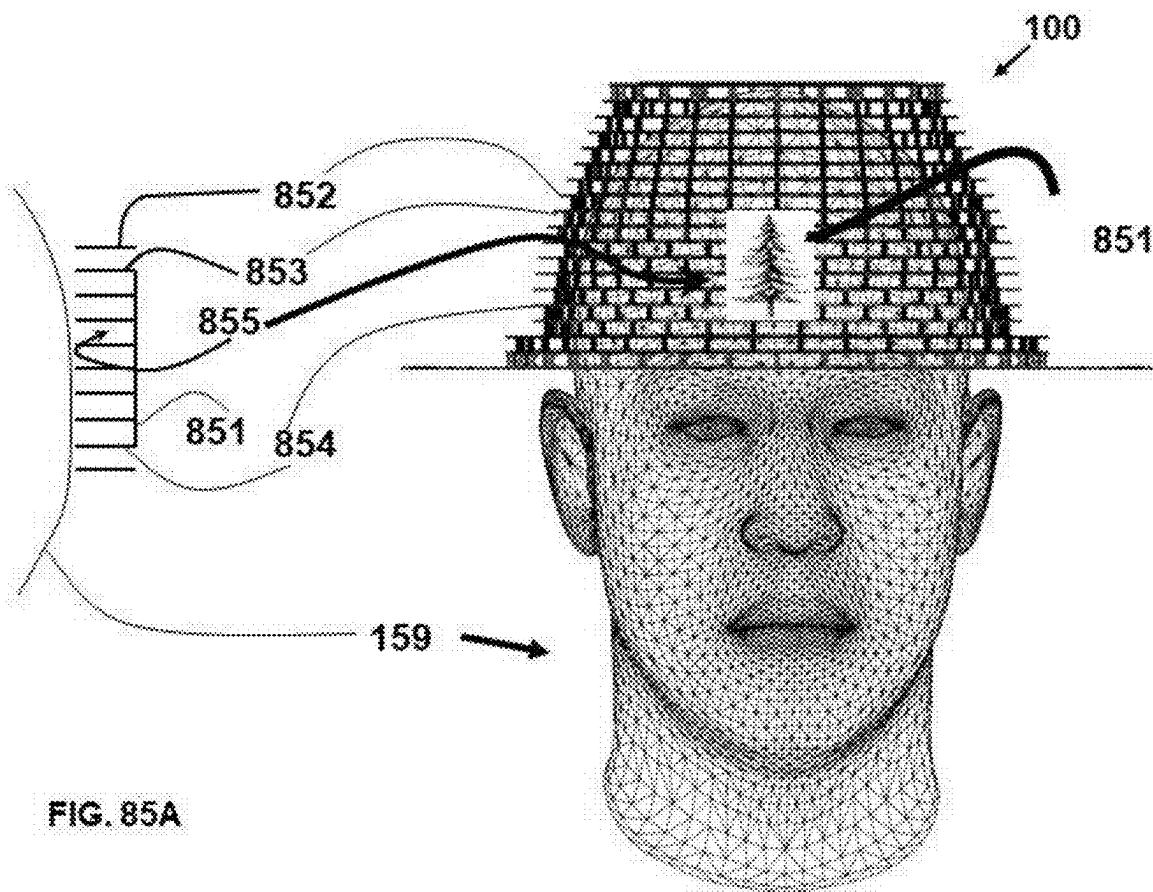


FIG. 85A

FIG. 85B

FIG. 85

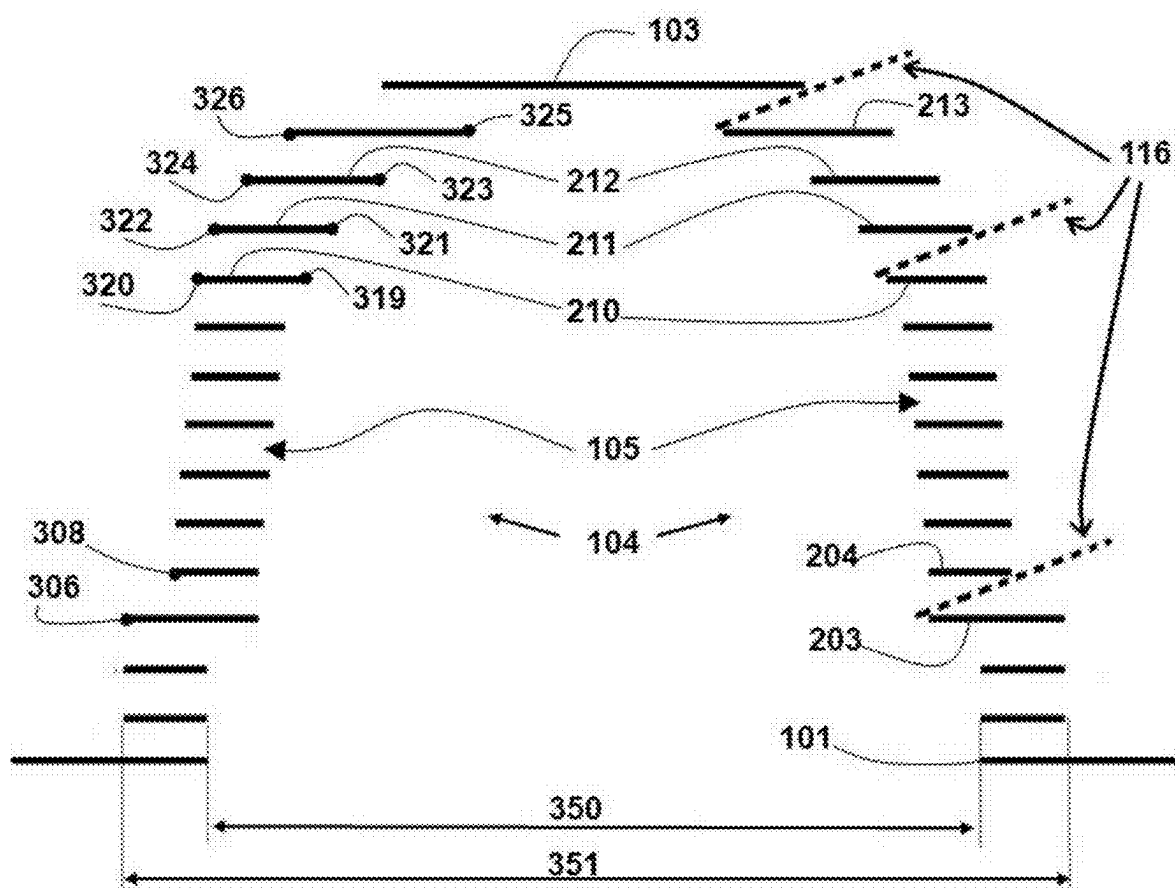


FIG. 110

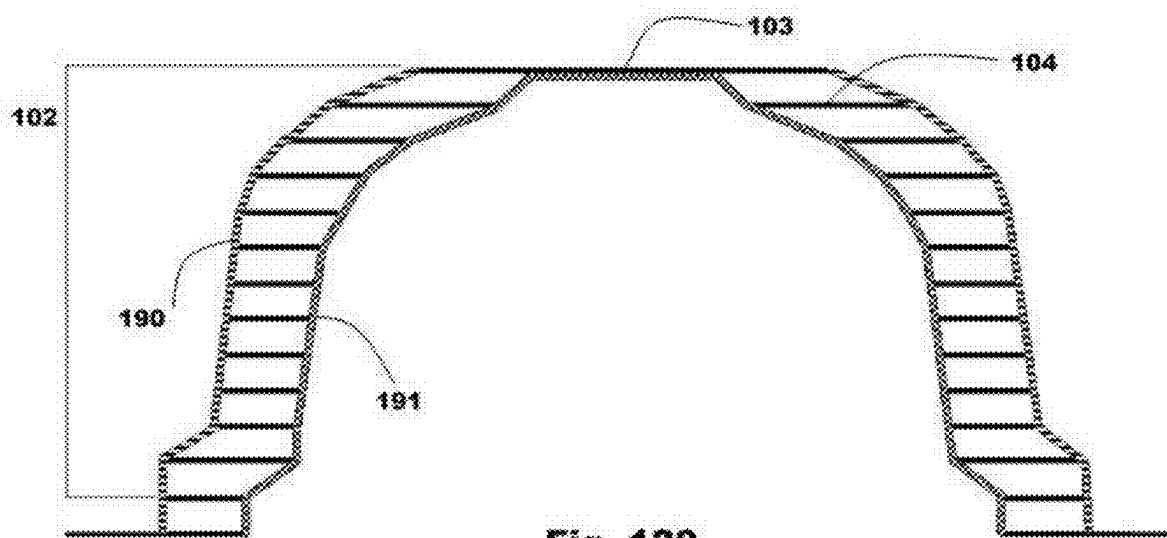
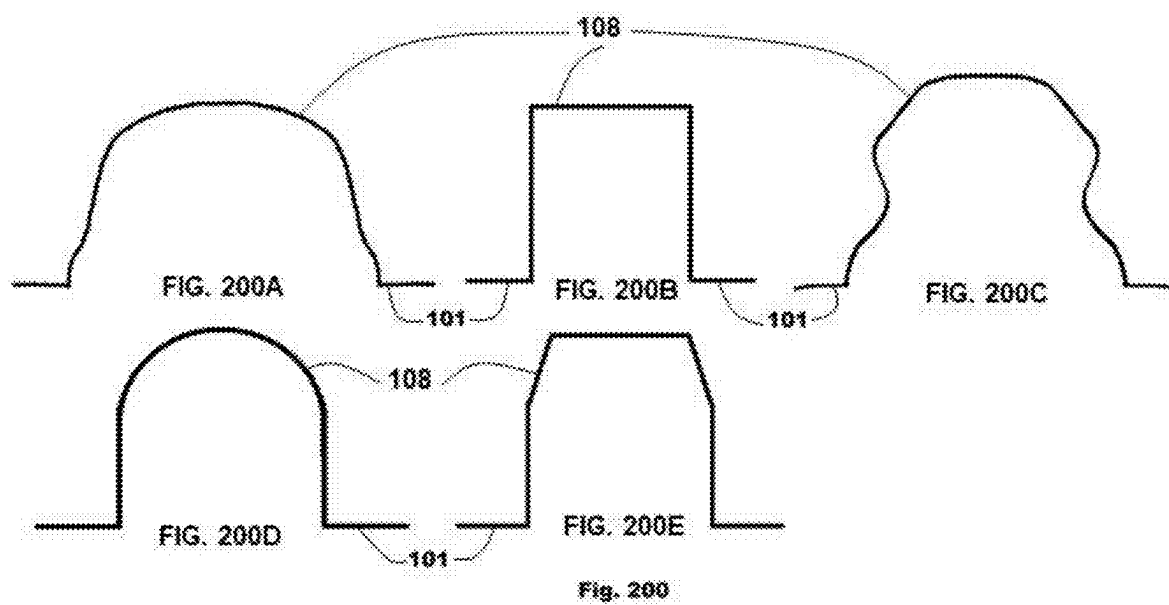


Fig. 120



VENTILATED HEADWEAR WITH LOUVERED CONSTRUCTION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/563,897, filed [Mar. 12, 2024], titled “Article of Headwear”, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] Numerous types of headwear have been used to shield against sunlight and improve visibility during outdoor activities. Traditional hats designed for shading the crown of the head trap heat and humidity, while those prioritizing ventilation tend to permit sunlight to reach the head. Therefore, an improved headwear is desired.

SUMMARY OF THE INVENTION

[0003] The present invention provides an article of headwear comprising a body having a plurality of horizontally-arranged, vertically-spaced louvers, spacers separating adjacent louvers, and a top member which can provide shade against sunlight while also prioritizing ventilation.

[0004] Each louver is formed as a generally annular ring having a first width. The louvers are spaced apart by a plurality of spacers disposed therebetween. The spacers space one louver from another by a first distance. The ratio of the first distance to the first width defines an effective spacing between the annular rings that prevents rays of the sun from impinging on the portion of the head of the wearer when an elevation angle of the rays is greater than a predetermined angle with respect to a horizontal ground plane.

[0005] This louvered construction allows for unimpeded airflow around the head, providing significant cooling compared to conventional headwear. The headwear according to the disclosure can advantageously be manufactured using additive or subtractive manufacturing techniques. The louvered construction can also be used as a component within a larger hat structure, providing ventilation to a portion of the hat.

BRIEF DESCRIPTION OF DRAWINGS

[0006] The features of the exemplary embodiments of the present invention will be described with reference to the following drawings, where like elements are labeled similarly, and in which:

[0007] FIG. 10A is a perspective view of a first example hat according to the disclosure;

[0008] FIG. 10B is a perspective view of a second example hat according to the disclosure;

[0009] FIG. 12 is a perspective view of a simplified version of the hat showing schematically sunlight blockage and free airflow;

[0010] FIG. 15A is a perspective view of the hat on a head viewed from 0 degrees.

[0011] FIG. 15B is a perspective view of the hat on a head viewed from 35 degrees above the head;

[0012] FIG. 20A is an oval shaped louver viewed from above;

[0013] FIG. 20B is a rectangular shaped louver viewed from above;

[0014] FIG. 30A is a perspective view of a flat louver ring laying on a grid ground plane;

[0015] FIG. 30B is a perspective view of a louver ring with a tilt angle;

[0016] FIG. 30C is a perspective view of a curved louver ring;

[0017] FIG. 50A is a schematic showing a configuration of louvers to block light at a specific angle;

[0018] FIG. 50B is a schematic showing a configuration of louvers to block light at a specific angle;

[0019] FIG. 50C is a schematic showing a configuration of louvers to block light at a specific angle;

[0020] FIG. 50D is a schematic showing a configuration of louvers to block light at a specific angle;

[0021] FIG. 50E is a schematic showing a configuration of louvers to block light at a specific angle;

[0022] FIG. 50F is a schematic showing a configuration of louvers to block light at a specific angle;

[0023] FIG. 50G is a perspective drawing of a vertical slice of a hat showing the construction of the edges of the hat;

[0024] FIG. 60A a perspective drawing of post type spacer;

[0025] FIG. 60B a perspective drawing of a post type spacer between two louvers;

[0026] FIG. 60C a schematic drawing of a post type spacer between two louvers;

[0027] FIG. 60D a perspective drawing of a spiral spacer;

[0028] FIG. 60E a perspective drawing of spiral spacers disposed between 3 louvers;

[0029] FIG. 60F a perspective drawing of a spacer formed of springy filament formed as a sphere;

[0030] FIG. 80A a top view perspective drawing of a person's head with schematic arrows indicating airflow;

[0031] FIG. 80B a perspective drawing of a person wearing the hat with a horizontal slice removed;

[0032] FIG. 80C a top view of the slice from FIG. 80B with schematic arrows indicating airflow;

[0033] FIG. 85A a cross section schematic showing a head, louvers and a placard attached to the louvers;

[0034] FIG. 85B a perspective front view of a person wearing the hat with a placard on the front of the hat;

[0035] FIG. 110 is a vertical schematic cut away view of the louvers in a hat;

[0036] FIG. 120 is a vertical schematic cut away view of the louvers in a hat supported by an inner and outer frame;

[0037] FIG. 200A is a vertical profile of a shape of a hat crown;

[0038] FIG. 200B is a vertical profile of a shape of a hat crown;

[0039] FIG. 200C is a vertical profile of a shape of a hat crown;

[0040] FIG. 200D is a vertical profile of a shape of a hat crown; and

[0041] FIG. 200E is a vertical profile of a shape of a hat crown.

[0042] All drawings are schematic and not necessarily to scale. Parts given a reference numerical designation in one figure may be considered to be the same parts where they appear in other figures without a numerical designation for brevity unless specifically labeled with a different part number and described herein.

DETAILED DESCRIPTION OF THE INVENTION

General Considerations

[0043] The features and benefits of the invention are illustrated and described herein by reference to exemplary embodiments. This description of exemplary embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. Accordingly, the disclosure expressly should not be limited to such exemplary embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features.

[0044] In the description of embodiments disclosed herein, unless explicitly indicated otherwise, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Unless explicitly indicated otherwise, relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top” and “bottom” as well as derivative thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation.

Definitions

Unimpeded

[0045] As used herein, the term unimpeded often used with “airflow” refers to airflow that is not significantly restricted or blocked by barriers. It is understood that some minimal air resistance will occur due to the surfaces of the louvers themselves and the presence of spacers, but this minimal resistance does not constitute significant impediment within the meaning of this invention. The airflow is considered “unimpeded” if it is substantially free to move in all horizontal directions and angles for example radial, tangential, and oblique directions around the wearer’s head.

Hat

[0046] As used herein, the term “hat” refers broadly to any article of headwear designed to be worn on or affixed to the head. This includes, but is not limited to, hats, caps, helmets, hoods, visors, and other forms of protective, ornamental, or functional headgear. The term encompasses headwear of various shapes, sizes, and materials, regardless of specific design, purpose, or intended use. The term “hat” may also include configurations that incorporate additional elements, such as ventilation systems, shading structures, or protective components, and is not limited to traditional or commonly understood definitions of the word.

Generally Ring Shaped

[0047] As used herein, the term ‘generally ring shaped’ refers to louvers that form a closed or substantially closed loop around the wearer’s head. This includes, but is not limited to, circular, oval, elliptical, rounded rectangular, rounded polygonal, and irregular shapes that enclose the head. The term does not require perfect geometric regularity.

Flat, Tilted, and Curved Rings/Louvers

[0048] FIG. 30A, 30B, 30C illustrate three exemplary louver configurations in perspective view: a flat louver **310**, a tilted louver **311**, and a curved louver **312**, respectively. Each louver is depicted as ovoid or generally circular. A reference grid **320** is included to indicate a horizontal plane.

[0049] Louver **310** is a flat louver, as defined herein. The inner edge **325** and the outer edge **326** of the louver lie substantially within the same plane, indicated by the reference grid **320**.

[0050] Louver **312** is a tilted louver, as defined herein. The inner edge **325** is positioned at a higher elevation relative to the reference grid **320** than the outer edge **326**.

[0051] Louver **303** is a curved louver, as defined herein. The inner edge **325** is positioned at a higher elevation relative to the reference grid **320** than the outer edge **326**.

Horizontal

[0052] As used herein, the term “horizontal plane” refers to a plane substantially parallel to an idealized, locally flat portion of the Earth’s surface, disregarding topographical variations.

Target Shade Angle

[0053] The hat is designed to block direct sunlight above a specific angle. As used herein, the term “shade angle” refers to the angle, measured relative to a horizontal plane, above which the louvered structure of the hat is designed to substantially block direct sunlight from reaching the wearer’s head.

Introduction

[0054] FIG. 10A, 10B depicts two exemplary embodiments of articles of headwear **100** according to the present disclosure. Unlike traditional mesh or screen hats, which provide ventilation at the expense of shading the head, and unlike sun blocking materials which provide shade, but block airflow and trap heat; the article of headwear **100** allows for significant ventilation in all horizontal directions without compromising shade through the use of louvers that comprise the body. The louvers block sunlight and provide shade to a user while also allowing for air to flow through the gaps between the louvers, thus providing ventilation to keep a user cool. The louvers may be configured to block sunlight from above a target shade angle.

Horizontal Ventilation with Angled Light Occlusion

[0055] The invention leverages the fact that the most significant heating effect from direct sunlight occurs when the sun is high in the sky. Conversely, airflow along the surface of the earth is generally horizontal. Even in the absence of a breeze, a person moving about will create an effective horizontal airflow.

[0056] FIG. 12 illustrates how the louvers block direct sunlight from higher angles **121** from entering the hat, partially blocks light from intermediate angles **122** and **123** from entering the hat. It also illustrates how horizontal air movement is not blocked by the hat.

[0057] FIG. 15A is a drawing taken from an actual CAD design showing how the hat does not block horizontal access to airflow to a head. FIG. 15B shows how the sun can’t “see” the head and therefore its rays don’t strike the wearer’s head.

Temperature Reduction Testing

[0058] Comparative testing demonstrates that the headwear described herein provides a significant reduction in temperature compared to conventional hat designs. In testing, a container of water was placed beneath the inventive headwear and various control hats, including baseball caps and safari hats, both with and without mesh ventilation panels. All hats were exposed to direct sunlight for a period of one hour. Subsequent temperature measurements of the water in each bowl revealed that the water under the inventive headwear was consistently 5 to 20 degrees Fahrenheit cooler than the water under the control hats.

Overall Structure and Components

[0059] Referring to FIG. 10, the article of headwear 100 may include a brim 101, and includes a body 102, and a top member 103. The body 102 further includes a plurality of horizontally-arranged, vertically-spaced louvers 104 separated by spacers 106.

[0060] The plurality of louvers 104 act as layers that, in combination with the spacers 106 and top member 103, comprise the body 102. Each layer of louvers 104 may include different enclosing shapes and sizes depending on the style of headwear. For example, in some embodiments, as shown for example in FIG. 20, the louvers 104 may be an enclosing shape and may be oval FIG. 20A, circular, rectangular FIG. 20B, square, or any shape to house a user's head within the body 102.

[0061] In some embodiments, as shown for example in FIG. 110, louvers have an interior radius 350 and an exterior radius 351. Note louver 210 may have an interior radius 319 and an exterior radius 320 which is greater than 319. Going further up the louvers towards the top member 103, louver layer 211 may have an interior radius 321 and an exterior radius 322, and so on (323 and 324, 325 and 326 . . .) for each layer of louvers until reaching the top member 103.

[0062] In some embodiments, the interior radius 350 of the louvers have a minimum radius of that of a user's head. In some embodiments, the minimum radius may vary from hat to hat to accommodate different head sizes.

[0063] Further, in some embodiments, the exterior radii (306, 308 . . .) may be modified to create any desired shape and overall exterior design of the article of headwear. Variations in the overall shape of the hat body, both in the shape of the individual louvers FIG. 20 and in vertical profile FIG. 200, require corresponding adjustments to louver dimensions and spacing to maintain the target shade angle FIG. 50 item 59. The size, spacing, positioning and curvature of the louvers can be adjusted to create a wide range of shapes. These variations demonstrate the versatility of the louvered design and its ability to accommodate different aesthetic preferences and functional requirements while still providing the core benefits of shade and ventilation.

[0064] While the embodiments described herein primarily focus on hats where the louvered construction forms the majority or entirety of the hat body, it is also contemplated that the louvered construction of the present invention can be used as a component within a larger, more conventionally styled hat. For example, a hat maker could incorporate the louvered section described herein into the lower part of a crown of a hat, providing ventilation to that specific area, while using other materials and construction techniques for

the upper crown, top, or other portions of the hat. This allows for the combination of the superior ventilation benefits of the louvered construction with a wider range of aesthetic designs. It has come to the inventor's attention that hat makers are interested in incorporating the present invention in this manner. This modular approach expands the potential applications of the invention and allows for greater design flexibility.

[0065] The size and configuration of the layers of louvers 104, provide shading from sunlight. Each louver provides shading to the louver below it. For example, as shown in FIG. 110, layer 103 blocks the sunlight 116 from entering the body at layer 213, and layer 211 blocks the sunlight from entering the body at layer 210, layer 204 blocks the sunlight from entering the body at layer 203.

Target Shade Angle

[0066] This design of the louvers allows for blocking of sunlight at or above a target shade angle 59, as shown in more detail in FIG. 50. The shade angle is the angle created between the outer edge 51 of an upper louver 510 and the inner edge 52 of the adjacent lower louver 520. Depending on the configuration of the upper louver layer 510 in relation to the lower louver layer 520, the shade angle 59 will block more or less sunlight. Therefore, the lower the shade angle 59, the less sunlight 55 will be able to penetrate the hat body 102. Thus, a lower shade angle promotes a cooler hat for a user, since less sunlight will radiate through the hat onto the user's head. The shade angle in simpler cases generally may be calculated: $\text{angle} = \arcsin(\text{spacer height} / \text{louver blade width})$.

[0067] For louvers that are directly above each other FIG. 50A the target shade angle may be maintained with louvers that are the same width 57. When the louvers are part of the body that is curved FIG. 50C simply shifting the louvers does not maintain the defined shade angle 59. To maintain the target shade angle the positions and or sizes must be altered. In FIG. 50B the lower louver width 56 is increased to maintain the target shade angle 59. In FIG. 50D the vertical distance between the upper and lower louvers is decreased to maintain the angle 59. The target shade angle may be selected depending on the desired balance between shading, louver width, number of louvers, weight, and louver spacing. In some embodiments, the shade angle may vary between each layer of louvers.

[0068] In order to decrease the shade angle, the spacing between the louver layers needs to decrease FIG. 50D, or there needs to be a greater difference in interior radius 58 of the bottom layer in comparison to the exterior radius of the higher layer of louver FIG. 50H. In other embodiments, the shade angle can be modified through the thickening or widening of the louver layers.

Unimpeded Airflow

[0069] FIG. 80A is a top view of a head with schematic large and small arrows showing airflow around a head 159. FIG. 80C is a top view of a slice from a CAD drawing of a hat I have produced also showing airflow around a head 159. FIG. 80B shows where the slice came from. Items 81, 82 and 83 represent tangential, oblique, and perpendicular airflow towards the head. The small arrows represent secondary airflow after the air impacts the head. It can also be appre-

ciated from the drawing that airflow from any direction relative to the head is equally unrestricted.

[0070] As can be seen, the spacers **87** are small and sparsely distributed, presenting negligible resistance to airflow from any horizontal direction. This cross-sectional view is representative of the majority of the hat's vertical extent, as the louvers are thin relative to the height of the spacers. Consequently, the airflow around the head **159** in FIG. **80A** is substantially the same as the unobstructed airflow around the bare head depicted in FIG. **80C**.

[0071] The gaps between the louvers (FIG. **10** item **105**, FIG. **110** item **105**) facilitate this free airflow. This unimpeded airflow promotes ventilation and cooling of the user's head **159** by allowing the dissipation of heated and/or humidified air and enabling evaporative cooling. The louvered construction thus provides shade while virtually eliminating airflow resistance.

Comparison to Mesh

[0072] While mesh materials are sometimes incorporated into hat crowns to provide ventilation, they offer a significantly less effective solution than the louvered structure of the present invention. Mesh, by its nature, permits substantial direct light penetration, diminishing the shade provided to the wearer. Furthermore, mesh presents a non-negligible impediment to airflow, deflecting and disrupting airflow rather than allowing the substantially unimpeded flow achieved by the louvered configuration.

[0073] Furthermore, a critical distinction arises when considering the application of panels. These panels, which may be used for displaying logos, team affiliations, decorative artwork, or other indicia, are often attached to headwear. In mesh headwear, the application of a substantially planar panel directly to the mesh surface obstructs airflow through the covered portion of the mesh, effectively negating the ventilation benefits in that area. However, in the present invention, as depicted in FIG. **85**, the attachment of a panel **851** to the outer edges of the louvers **104** does not preclude substantial airflow. The inherent spacing and arrangement of the louvers create channels **855** between the panel and the wearer's head, permitting continued tangential and oblique airflow even in the region behind the panel. This maintained airflow contributes to the overall cooling effect of the headwear, even in areas where a panel is present.

Horizontal, Flat, Tilted and Curved Louvers

[0074] In some embodiments, the louvers **104** are flat members extending linearly along a single plane. In other embodiments, the louvers **104** may be curved from their innermost to outermost edges and/or tilted downwardly towards the adjacent lower louver.

[0075] FIG. **50E** and FIG. **50F** demonstrate how curved or tilted louvers can achieve a lower shade angle **59** without increasing the vertical spacing or the width **57** of the louvers. This design offers a potential advantage by allowing for a reduced hat body thickness or lower shade angle. However, there is a tradeoff in that tilting or curving the louvers can slightly impede airflow compared to perfectly horizontal louvers.

[0076] In some embodiments, the louvers may be adjustable, allowing the wearer to modify the shade angle or airflow characteristics of the headwear. This adjustability can be achieved through various mechanisms, such as

hinges, pivots, sliding tracks, or other suitable means. The specific implementation of adjustable louvers can vary depending on the design and intended use of the headwear. For example, in a sports helmet, adjustable louvers could allow the user to fine-tune ventilation based on activity level and weather conditions. In a fashion hat, adjustable louvers could provide a way to customize the appearance and shading effect.

[0077] This specification may contain references to "horizontally arranged" and "flat" louvers or similar language. Also, most figures depict idealized flat, horizontal louvers for clarity. However, throughout this specification, such references and depictions should be understood to encompass louvers that are tilted or curved, as discussed above and illustrated in FIGS. **55A**, **B**, **C**, **D** and **H**. Also louvers in this specification portray each louver being a single object that completely encloses the head. Again this is for clarity but it is equally possible to make louvers that are not continuous but collections of disconnected sections.

[0078] In some embodiments, the louvers **104** may be comprised of a plurality of circumferentially arranged louver segments. These segments, when assembled, collectively function to create the louvered structure of the hat body **102**, providing the desired shading and airflow.

Spacers

[0079] Spacers (FIG. **12** item **106**) are located between adjacent louvers **104**, extending from a first louver to an adjacent second louver. The height of the spacers determines the vertical spacing between the louvers. This spacing, in conjunction with the dimensions of the louvers, directly affects the shade angle **59**. Increasing the spacer height, while maintaining the louver dimensions, will generally increase the shade angle. Conversely, decreasing the spacer height (FIG. **50D** item **540**) will generally decrease the shade angle. To maintain a target shade angle while altering the spacer height, the dimensions of the louvers (specifically the difference between the inner and outer radii) must be adjusted accordingly. The vertical spacing may be uniform throughout the hat body **102**, or it may vary between different louver layers. The specific dimensions and spacing are chosen to achieve the desired balance between shading, airflow, overall hat size and appearance.

[0080] In some embodiments, as shown for example in FIG. **12**, the spacers **106** are spool like structures having a thin center rod that extends between each of the above and below layers. See also FIG. **60A**. Depending on the desired hat structure, spacers may be flexible or when more support is needed, spacers may be rigid. Spacers **106** extend circumferentially around the body **102** and connect two layers of louvers **104**, while also allowing unfettered air flow and minimizing air resistance in all directions to the article of headwear **100**, such as airflow incoming from the front, back, side, perpendicular, oblique, and tangential directions. For example, spacers **106** may have any of the configurations shown in FIG. **60**, however, the invention should not be limited to the embodiments shown. A plurality of individual spacers distributed around the louver is one way to support the louvers. FIG. **60F**, springy balls, and FIG. **60A**, posts, show two such spacers.

[0081] The spacers **106** may take the shape of other configurations that support the layers of louvers **104** while also allowing unfettered air flow. For example, in some embodiments, the spacers will have a smooth, thin, and

round shape to minimize air resistance. However, although not optimal for air resistance, spacers **106** may have a rough texture and/or sharp edges that provide more aesthetic appeal. In some embodiments, 10-99% of the space between the layers of louvers (FIG. **110** item **105**) is open to air flow and is not occupied by the spacers **106**. Further, in some embodiment, 66% of the space between the layers of louvers (FIG. **110** item **105**) is open to air flow and is not occupied by spacers **106**. The best spacer designs disturb airflow minimally and the same in all directions. Wall shaped spacers would degrade performance.

[0082] In some embodiments, as shown for example in FIG. **60D**, spacers are elongated structures disposed along louvers and extending between adjacent louver layers, providing periodic support. FIG. **60E** shows spiral spacers **65** disposed between three louvers **104**. Prototype testing has demonstrated the effectiveness of this spiral spacer configuration.

[0083] The spacers will be smooth and round to minimize air resistance. However, although not optimal for air resistance, spacers may have a rough texture and/or sharp edges. In some embodiments, the article of headwear **100** will only have 2 support points per layer of louver to support each half of the louver, however in other embodiments, the article of headwear will include more support points at closer displacements throughout the layer of louvers **104** to provide greater support. For example, in uses such as a helmet, spacers may be displaced closer together than a recreation hat, such that the helmet can have more support and strength, to provide greater protection to the user.

[0084] Some embodiments utilize spacers that are collapsible or springy, to permit the hat to be partially collapsed and make it more durable. FIG. **60F** is a depiction of one such embodiment. It is a sphere made of flexible springy filament attached to adjacent louvers at **601**. In other embodiments a spring spiral aligned vertically may serve the same purpose.

[0085] In some embodiments, shown for example in FIG. **120**, the article of headwear **100** does not include any spacers, and instead utilizes an internal **191** and/or external frame **190** to support the louvers. The frames **190**, **191** may extend circumferentially around the louvers. In some embodiments, the frames **190**, **191** are porous allowing efficient airflow. In some other embodiments the frames **190**, **191** may run vertically from the top of the hat to the bottom creating a birdcage like structure to support the louvers. In some embodiments the frames may be of a wire like material which is round and smooth to minimize air resistance. However, although not optimal for air resistance, frame may have a rough texture and/or sharp edges.

Styles

[0086] In some embodiments, the article of headwear **100** may take the shape of a baseball cap FIG. **10B**, sunhat, helmet, sports hat, hood, or other article of headwear. Depending on the form, the elements of the article of headwear will vary accordingly. For example, as described above, in the example of the sports helmet, the louvers may be comprised of more rigid materials and the supports and spacers may be closer together to provide additional support and protection to the user. Further, in another example of the sports helmet, the louvers may include shock absorbing elements. In another embodiment, the article of headwear **100** may maximize ventilation and lightweight materials to optimize a user's experience in the heat.

Prototyping and Low-Volume Production

[0087] Initial prototypes of the headwear were constructed using a combination of subtractive and additive manufacturing techniques, demonstrating the feasibility of the louvered design. A custom software application was developed to facilitate the design and fabrication process.

Software:

[0088] A custom software application, integrated with a computer-aided design (CAD) program (Blender), was developed to generate three-dimensional (3D) models and manufacturing files for the headwear. This software takes user inputs from a spreadsheet and, at least one 2D vector drawing. The spreadsheet allows the user to specify key parameters for each louver, including:

- [0089]** Inner diameter
- [0090]** Outer diameter
- [0091]** Louver thickness
- [0092]** Spacer height (or spacing between louvers)
- [0093]** Selection of reinforcing ribs (optional)
- [0094]** Inclusion of a top member

[0095] The software automatically calculates and displays the resulting shade angle based on these parameters. The 2D vector drawing(s) define the enclosing shape of the louvers (e.g., oval, circular, rectangular, or other custom shapes). The software accepts vector drawings in standard formats (e.g., SVG, DXF).

[0096] Single Vector Drawing: If the hat body is to have a uniform louver shape from bottom to top (e.g., all circular louvers), a single vector drawing is provided. The software then uses the spreadsheet parameters to extrude this shape and create the stacked louver structure.

[0097] Multiple Vector Drawings: If the hat body is to have varying louver shapes (e.g., transitioning from an oval at the brim to a circle at the top), multiple vector drawings are provided, each representing the shape of a louver (or a group of louvers) at a specific height. The software interpolates, or transitions, between these shapes to create a smooth, continuous hat body. The spreadsheet parameters still control the dimensions and spacing, but the vector drawings define the plan view shape at different points along the hat's height.

[0098] The software can generate:

- [0099]** 3D models suitable for 3D printing (e.g., in STL format).
- [0100]** 2D cut files suitable for subtractive manufacturing (e.g., in SVG or DXF format).
- [0101]** 3D models for the jigs.

[0102] This software significantly streamlines the design process and allows for rapid iteration of different louver configurations and hat styles.

Subtractive Manufacturing (Cutting):

[0103] Louvers for prototypes were cut from sheet materials, including vinyl and acrylic, using a computer-controlled cutting machine (specifically, a Cricut Explore Air 2). The cut files generated by the custom software were used to control the cutting machine. This method proved effective for producing flat, horizontal louvers with high precision.

[0104] Spacers: Two types of spacers were prototyped:

[0105] Spool Spacers: Small, spool-shaped spacers (approximately 1.5 mm post diameter, 7 mm height, 3 mm flange diameter) were fabricated using stereolithography

(SLA) 3D printing with an ABS-like resin (Sunlu ABS-like). This provided the required precision for these small components.

[0106] Spiral Spacers: PVC spiral binder was commercially sourced.

Additive Manufacturing (3D Printing):

[0107] Full Hat Sections: Later prototypes involved 3D printing entire sections of the hat (e.g., halves) using SLA printing, demonstrating the feasibility of producing the complete louvered structure with additive manufacturing.

Assembly:

[0108] Prototype assembly involved manually positioning the cut or 3D-printed louvers and spacers. A jig was used to ensure correct alignment and spacing during assembly. Adhesives (such as cyanoacrylate) were used to bond the components together.

Mass-Market Production

[0109] For high-volume production, thermoforming and injection molding are the preferred manufacturing methods. These methods offer advantages in terms of cost-effectiveness, production speed, and the ability to create complex shapes (including curved or tilted louvers). Subtractive methods such as using a laser cutter may be suitable for flat louvers.

Thermoforming

[0110] Individual louvers can be efficiently produced via thermoforming, particularly vacuum forming. A sheet of thermoplastic material (such as HDPE, LLDPE, or PETG) is heated and drawn over a mold defining the louver's shape. The mold incorporates the desired curvature, tilt (if any), and inner/outer edge profiles, and is typically made of aluminum, wood or 3d printed resin. Multiple louvers can be formed simultaneously from a single sheet using a multi-cavity mold. After forming and cooling, the louvers are trimmed from the sheet using methods such as CNC routing, laser cutting. The thickness of the sheet is selected to match the target louver thickness (e.g., 0.5 mm-1 mm).

Injection Molding

[0111] Injection molding offers a highly efficient method for mass-producing the louvers and, optionally, the spacers. Due to the thin cross-section of the louvers, thin-wall injection molding techniques are employed:

[0112] Material Selection: High-flow polymers like HDPE and LLDPE are preferred for their flow properties, strength, and UV resistance.

[0113] Rapid Injection and Heated Molds: High injection speeds and heated molds ensure complete filling of the thin louver cavities. Precise temperature control is critical.

[0114] Sectional Construction: The hat body is preferably molded in sections (e.g., quadrants) to reduce mold size, improve flow control, and simplify demolding. Sections are designed with interlocking features (snap-fit connections, overlapping joints) for assembly.

[0115] Optimized Mold Design: Molds incorporate multiple gates, venting, and cooling channels.

[0116] Overmolding: Spacers can be overmolded onto the louvers in a two-step process, eliminating separate spacer

attachment. In a two step process the louvers would be made, and then they would be placed in a second mold for the spacers to be created. The material used for the spacers can be the same or different from the material used for the louvers.

Assembly

[0117] Assembly of the louvered hat, regardless of the louver production method, involves the following considerations:

[0118] Jig-Assisted Manual Assembly: A jig is essential for accurate and efficient manual assembly, holding louvers in position while spacers are attached.

[0119] Adhesive Bonding: A suitable adhesive (cyanoacrylate, epoxy, UV-curing, or solvent-based, depending on materials) is used to secure spacers to louvers.

[0120] Automated Assembly (Discrete Spacers): Pick-and-place robots with small-part grippers and vibratory bowl feeders can automate spacer placement and bonding for higher volumes. A vision system may enhance accuracy.

[0121] Alternative Attachment: Heat-activated adhesives, applied to the spacers or louvers, offer another potential assembly method.

[0122] Spiral Spacer (Manual Assembly) A pre-formed, continuous spiral spacer (made from a material similar to heavy-gauge fishing line, potentially with a non-circular cross-section for added stiffness) can be manually inserted between louvers, aided by the assembly jig. While automated assembly methods are suitable for high-volume production, the use of a spiral spacer in conjunction with a simple jig enables a highly efficient, low-capital manual assembly process. This is particularly advantageous for initial market entry, low-volume production runs, or situations where capital investment in automation is not yet justified.

Low-Capital, Scalable Production Using Manual Assembly

[0123] A significant advantage of the louvered headwear design is its suitability for low-capital, scalable production using manual assembly techniques, particularly with the spiral spacer method. Spiral spacers may not be the best mass market type of spacers, but they may serve as an excellent low cost intermediate step. As described in the Assembly section, the combination of pre-cut or thermoformed louvers, a readily available spiral spacer, and a simple assembly jig allows for efficient manual assembly. Employing rapid bonding methods, such as a pre-coated spiral with heat-activated or UV-curing adhesive, further enhances production speed. This approach enables a manufacturer to begin producing the headwear with minimal initial investment in tooling or automation, making it ideal for startups, small businesses, or initial market testing. Production volume can be readily scaled up by adding additional assembly jigs and personnel, or by transitioning to semi-automated or fully automated methods as demand increases. This low-barrier entry point and inherent scalability significantly enhance the commercial attractiveness of the invention.

CONCLUSION

[0124] While the foregoing description and drawings represent exemplary embodiments of the present disclosure, it will be understood that various additions, modifications and

substitutions may be made therein without departing from the spirit and scope and range of equivalents of the accompanying claims. In particular, it will be clear to those skilled in the art that the present invention may be embodied in other forms, structures, arrangements, proportions, sizes, and with other elements, materials, and components, without departing from the spirit or essential characteristics thereof. In addition, numerous variations in the methods/processes described herein may be made within the scope of the present disclosure. One skilled in the art will further appreciate that the embodiments may be used with many modifications of structure, arrangement, proportions, sizes, materials, and components and otherwise, used in the practice of the disclosure, which are particularly adapted to specific environments and operative requirements without departing from the principles described herein. The presently disclosed embodiments are therefore to be considered in all respects as illustrative and not restrictive. The appended claims should be construed broadly, to include other variants and embodiments of the disclosure, which may be made by those skilled in the art without departing from the scope and range of equivalents.

I claim:

1. A hat, comprising:
 - a cavity for receiving a user's head;
 - a plurality of generally ring shaped vertically spaced louvers forming at least a portion of a hat body configured to at least partially surround a wearer's head;
 - the louvers are configured to block passage of rays of the sun between the louvers when the rays are incident on a vertical plane of the hat at angles greater than a target shade angle with respect to a ground plane regardless of variations in shape of the hat body;
 - the louvers being arranged to allow substantially unimpeded airflow around the wearer's head; and
 - and wherein the hat body comprises at least one louver that is not horizontally congruent with another louver.
2. The hat of claim 1, further comprising a top member that prevents sunlight from entering the top of the hat body.
3. The hat of claim 1, wherein the target shade angle is less than about 70 degrees.
4. The hat of claim 1, wherein the louvers are substantially flat.
5. The hat of claim 1, wherein at least one louver is connected to a spacer, and wherein a plane defined by said at least one louver is not parallel to a horizontal plane.
6. The hat of claim 1, wherein the louvers are adjustable to vary an amount of light blocked.

7. The hat of claim 1, wherein at least one louver is fragmented into a plurality of segments, and wherein adjacent segments of the fragmented louver are vertically offset from each other while maintaining the target shade angle defined by the louvers.

8. The hat of claim 1, wherein the spacers are in the form of discrete posts.

9. The hat of claim 1, wherein the spacers are configured to minimize airflow impedance regardless of the angle of incidence of the airflow relative to the spacer's surface.

10. A method of manufacturing an article of headwear, the method comprising:

forming a louvered structure comprising a plurality of horizontally-arranged, vertically-spaced louvers, wherein at least one louver has a different shape or size than another louver, and wherein adjacent louvers overlap vertically; and

assembling the louvered structure to form at least a portion of a hat body.

11. The method of claim 10, wherein forming the louvered structure comprises cutting a plurality of individual louvers from sheet material and attaching a plurality of discrete spacers between adjacent louvers.

12. The method of claim 10, wherein forming the louvered structure comprises thermoforming a plurality of individual louvers from a thermoplastic sheet and attaching a plurality of discrete spacers between adjacent louvers.

13. The method of claim 10, wherein forming the louvered structure comprises injection molding at least one section of the hat body, wherein the at least one section comprises a plurality of the louvers and a plurality of spacers integrally molded with the louvers.

14. The method of claim 13, wherein injection molding comprises injecting thermoplastic material into a mold at high speed and maintaining the mold at an elevated temperature.

15. The method of claim 13, wherein the hat body is formed in a plurality of sections, and wherein injection molding comprises molding each section separately.

16. The method of claim 15, further comprising joining the sections using snap-fit connections.

17. The method of claim 10, wherein forming the louvered structure comprises 3D printing.

18. The method of claim 10, wherein assembling comprises using a jig to align components of the hat body.

19. The method of any one of claim 11, 12, or 15, wherein assembling comprises using an adhesive to bond components of the hat body together.

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